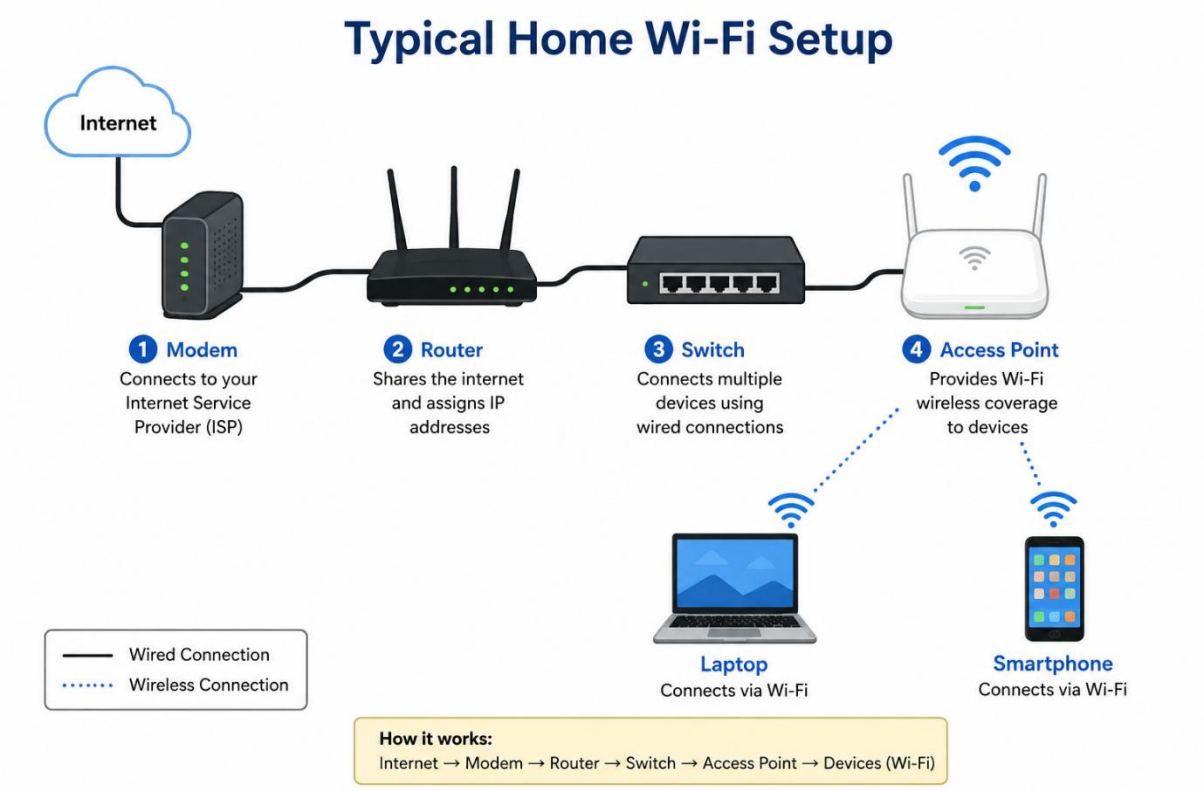
1.

Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.



1. Internet – Provides the internet connection.
2. Modem – Receives the internet from the ISP.
3. Router – Shares the internet with devices.
4. Switch – Connects multiple wired devices.
5. Access Point – Provides Wi-Fi connection.
6. Laptop – Connects to the network using Wi-Fi.
7. Smartphone – Connects to the network using Wi-Fi.

Connection:
Internet → Modem → Router → Switch → Access Point → Laptop & Smartphone

2.

List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it (for example, 'router: at home to share internet').

Device	Main Function	Real-Life Example
Modem	Brings internet to your home.	Used at home to get internet from the ISP.
Router	Shares internet with different devices.	Used at home to connect phones and laptops.
Switch	Connects many devices with cables.	Used in an office to connect computers.
Access Point	Gives Wi-Fi to devices.	Used in schools or offices for Wi-Fi.
Network Cable	Connects devices and carries data.	Ethernet cable connects a computer to a router.

3.

Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.







Cable Type	Where it is commonly used
1. Ethernet Cable (RJ45)	Used to connect computers, routers, switches and other devices in home/office LAN networks.
2. Coaxial Cable	Commonly used for cable TV, broadband internet, CCTV and antenna connections.
3. Fiber Optic Cable	Used for high-speed internet, data centers and long-distance communication because it carries data using light.

4.

Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data.

Difference	Router	Switch
Main use	Connects different networks and provides Internet.	Connects devices in the same network.
Location	Usually at home or office, connected to the Internet.	Usually inside offices, schools, or computer labs
Data handling	Sends data between different networks.	Sends data to the correct device in the same network.
Uses	Uses IP addresses to send data.	Uses MAC addresses to send data.
Example	Home Wi-Fi router.	Office network switch.